CHM_159
Worksheet_Chapter_1_Answers

Classifying matter, law of definite proportions, law of multiple proportions, elemental symbols, average atomic mass, the mole, molar mass

1. Classify the following matter as an element, compound, homogeneous mixture, or heterogeneous mixture.

(a) hydrogen, (b) water, (c) gold, (d) sugar.: (e) unfiltered seawater, (f) helium gas, (g) sodium chloride (salt), (h) a bottle of soft drink, (i) a milkshake, (j) air in a bottle, (k) concrete.

Ans: (**a) hydrogen: element**
(b) water: compound
(c) gold: element
(d) sugar: compound
(e)The sea is a heterogeneous mixture of seawater and biological matter, but seawater, with the biomass filtered out, is a homogeneous mixture
(f) element
(g) compound
(h) homogeneous mixture
(i) heterogeneous mixture
(j) homogeneous mixture
(k) heterogeneous mixture

2. A compound of copper and sulfur contains 88.39 g of metal and 44.61 g of nonmetal. How many grams of copper and sulfur are present in 5264 kg of the compound.

Ans : 3.48 x 10^6 g copper 1.766x10^6 g sulfur

3. A 55.5-g sample of water was decomposed into its constituent elements to produce 6.21 g hydrogen gas. Another sample of water was decomposed to produce 12.7 g oxygen gas. What is the mass of the 2nd sample of water?

Ans : 14.3 g

4. Compound X contains only carbon, hydrogen, and oxygen.

A 25.00-g sample of X contains 10.00 g of carbon.
A 50.00-g sample of X contains 26.64 g of oxygen.

What is the total mass of a sample of X that contains 8.06 g of hydrogen?

Ans : 120. g

5. The elements nitrogen and oxygen can form a variety of different compounds. Two such compounds, NO and N_2O_4, were decomposed into their constituent elements. One produced 0.8756 g N for every gram of O; the other produced 0.4378 g N for every gram of O. Show that these results are consistent with the law of multiple proportions.

Ans : $\frac{\text{ratio of N to O in compound 1}}{\text{ratio of N to O in compound 2}} = \frac{0.8756}{0.4378} \approx 2:1$

6. Fill in the blanks in the table:

Symbol		$^{54}_{26}Fe^{2+}$			
Protons	5			79	86
Neutrons	6		16	117	136
Electrons	5		18	79	
Net charge			-3		0

Ans :

Symbol	$^{11}_{5}B$	$^{54}_{26}Fe^{2+}$	$^{31}_{15}P^{3-}$	$^{196}_{79}Au$	$^{222}_{86}Rn$
Protons	**5**	**26**	**15**	**79**	**86**
Neutrons	**6**	**28**	**16**	**117**	**136**
Electrons	**5**	**24**	**18**	**79**	**86**
Net Charge	**0**	**+2**	**–3**	**0**	**0**

7. The atomic masses of ^{203}Tl and ^{205}Tl are 202.972320 and 204.974401 amu, respectively. Calculate the natural abundances of these two isotopes. The average atomic mass of thallium is 204.4 amu.

Ans : the natural abundances of ^{203}Tl and ^{205}Tl are 28.65% and 71.35%, respectively.

8. The atomic masses of ^{204}Pb (1.4 percent), ^{206}Pb (24.1 percent), ^{207}Pb (22.1 percent) and ^{208}Pb (52.4 percent) are 203.973020, 205.974440, 206.975872, and 207.976627 amu, respectively. Calculate the average atomic mass of lead. The percentages in parentheses denote the relative abundances.

Ans : 207.2 amu

9. Bromine has two naturally isotopes (Br-79 and Br-81) and has an atomic mass of 79.904 amu. The mass of Br-81 is 80.9163 amu and its natural abundance is 49.31 %. Calculate the mass and natural abundance of Br-79.

Ans 78.92 amu, 50.69 %

10. A hypothetical element (atomic mass = 12.34 amu) has three naturally occurring isotopes with isotopic masses and natural abundances given below. Calculate the percent abundance of Isotope 3

Isotope	Mass(amu)	Abundance(%)
1	11.09027	25.11
2	12.09894	----
3	16.13360	----

Ans : 12.25%

11. The following table gives numbers of electrons, protons, and neutrons in atoms or ions of a number of elements. Answer the following:

(a) Which of the species are neutral?

(b) Which are negatively charged?

(c) Which are positively charged?

(d) What are the conventional symbols for all the species?

		Atom or Ion of Element					
	A	**B**	**C**	**D**	**E**	**F**	**G**
Number of electrons	5	10	18	28	36	5	9
Number of protons	5	7	19	30	35	5	9
Number of neutrons	5	7	20	36	46	6	10

Ans : **(a)** Species with the same number of protons and electrons will be neutral. **A, F, G**.

(b) Species with more electrons than protons will have a negative charge. **B, E**.

(c) Species with more protons than electrons will have a positive charge. **C, D**.

(d) **A**: $^{10}_{5}\mathrm{B}$ **B**: $^{14}_{7}\mathrm{N}^{3-}$ **C**: $^{39}_{19}\mathrm{K}^{+}$ **D**: $^{66}_{30}\mathrm{Zn}^{2+}$ **E**: $^{81}_{35}\mathrm{Br}^{-}$ **F**: $^{11}_{5}\mathrm{B}$ **G**: $^{19}_{9}\mathrm{F}$

12a. What mass of magnesium , in g, is needed to provide 5.00×10^{6} atoms?

Ans: 2.02×10^{-16}g

12b. 868 mg of a certain element contains 2.61×10^{21} atoms. Identify the element by finding its molar mass.

Ans : Hg

13. A 2.4 ng sample of an unknown, neutral element is known to contain 1.20×10^{14} atoms. How many electrons does one atom of this element contain?

Ans: 6 electrons

14. The isotopic mass of silicon-30 is 29.973 amu. What amount of neutrons, in moles, is contained within a sample of silicon-30 having a mass of 68.7 g?

Ans: 36.7 moles of neutrons

15. A 1.10 mol sample of an element has a density of 6.506 g/cm^3. If the sample occupies a volume of 15.42 cm^3, what is the molar mass of the element?

Ans : 91.202 g/mol

16. The following charges on individual oil droplets were obtained during an experiment similar to Millikan's. Use them to determine a charge for the electron in coulombs (C), showing all your work.

Charges (C): -3.184×10^{-19}; -4.776×10^{-19}; -7.960×10^{-19}

Ans : -1.592×10^{-19}C